

MedEvi-NS at ArchEHR-QA 2026: *Teaching LLMs to think like clinicians*

Using Clinical Reasoning Principles to Improve Zero-shot Capabilities of Large Language Models in Evidence Alignment

Mengxuan Sun¹ and Nicolay Rusnachenko²

¹University of Aberdeen, ²Bournemouth University

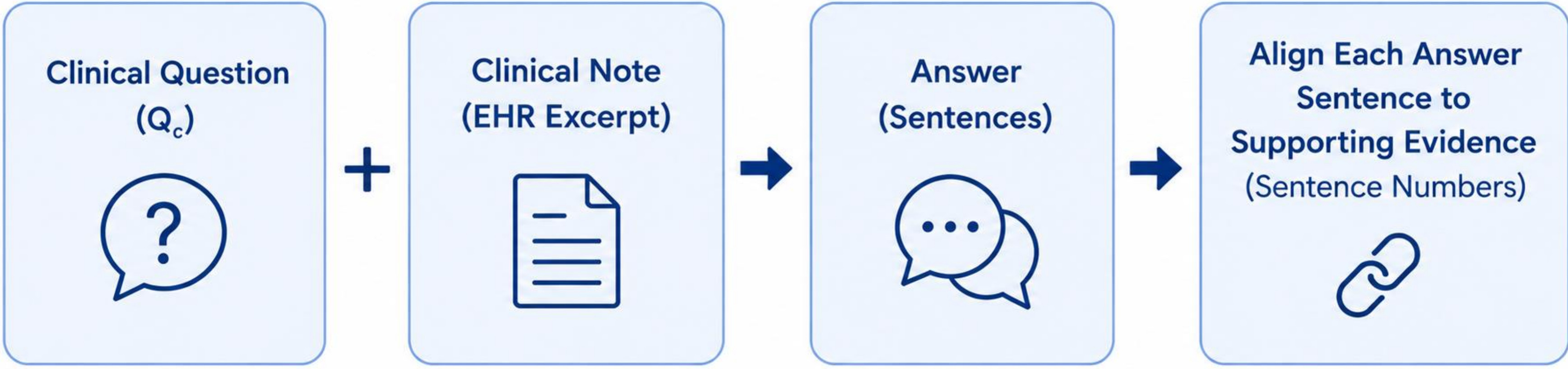


The Problem

LLMs may generate answers that **sound plausible but are not truly supported by clinical evidence.**

In medicine, answers must be **grounded in real patients' data.**

The Task (ArchEHR-QA Task 4)



Goal: For each sentence in the answer, find the exact supporting sentence(s) in the clinical note.

Which note matches the answer?

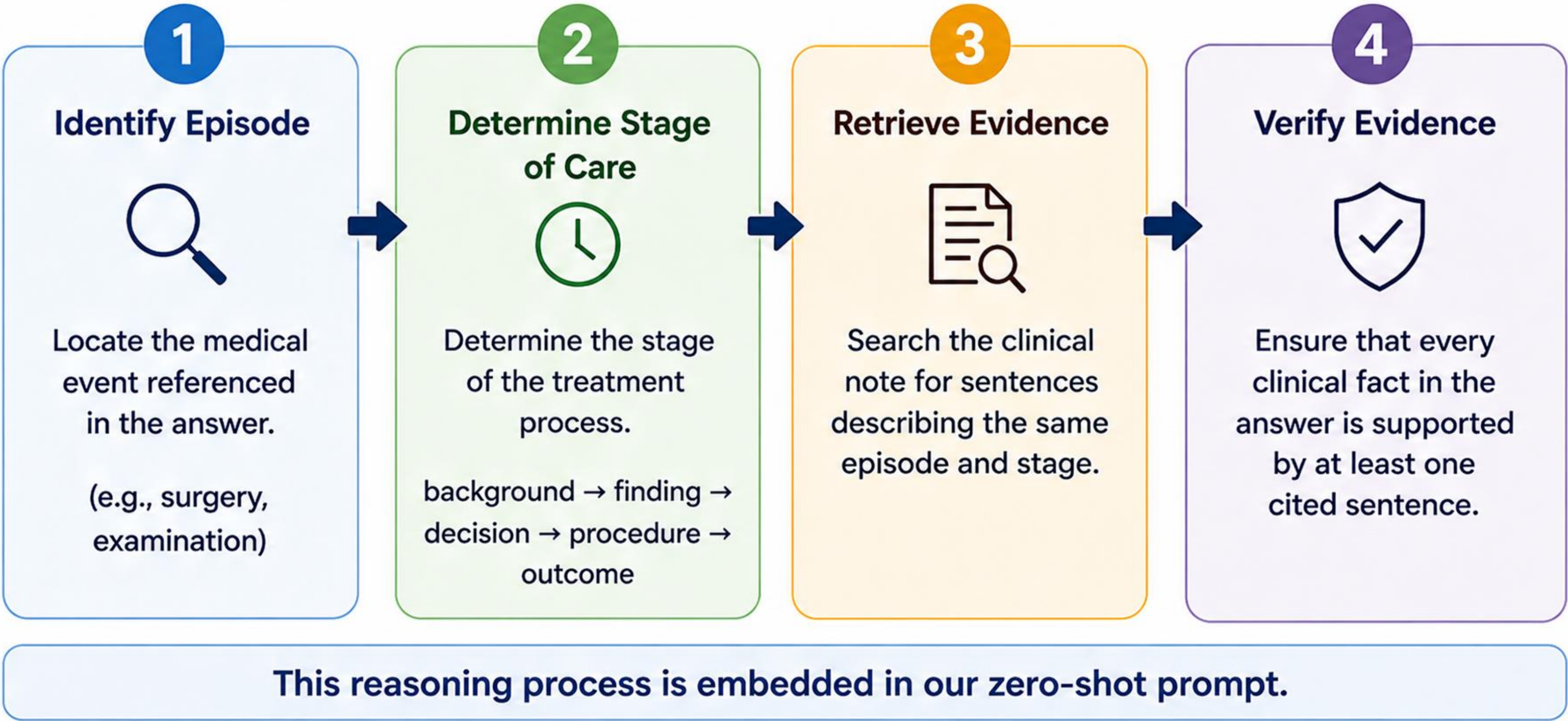
- Note 1:
- High keywords overlap
 - This is a **recommendation**
 - Different stage: **Decision**

- Clinical Note**
1. The doctor *recommended* ERCP for the patient.
 2. ERCP *was performed* and stones removed

Answer
👉 The patient underwent ERCP

- Note 2:
- Less exact wording
 - This **described** what happened
 - Same stage: **Procedure**

Our Clinical Reasoning Framework



Experiment Settings

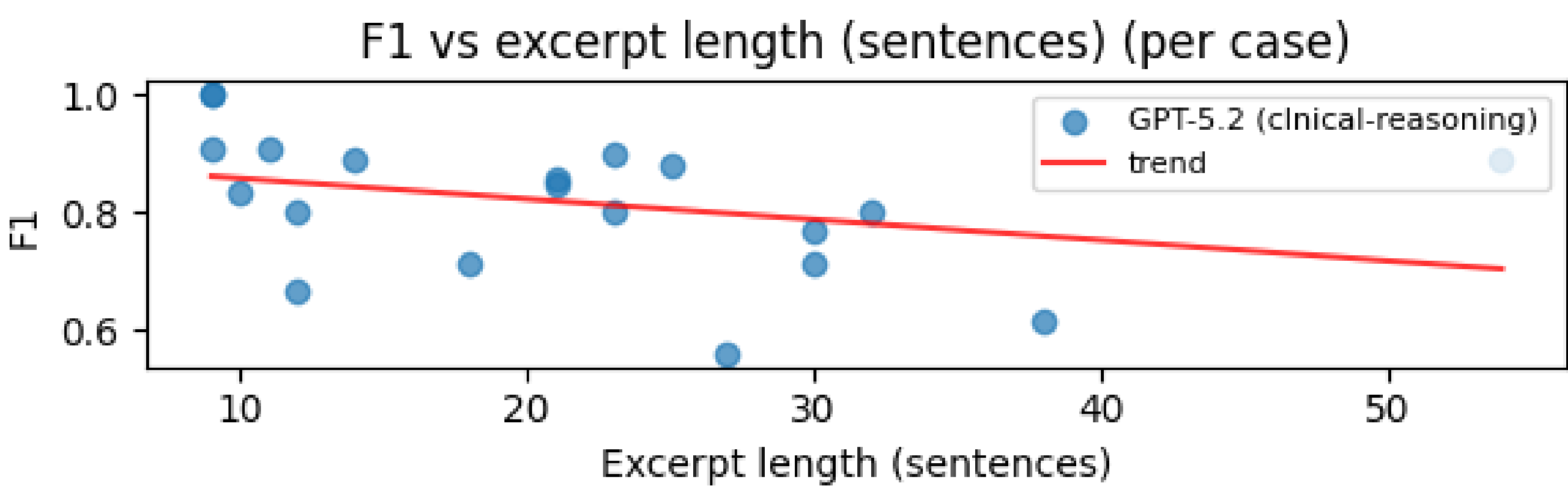
Dataset: ArchEHR-QA(v1.5)

- Avg 28 sentences per note
- Avg 4-5 sentences per answer
- Avg 102 characters per sentence

Model: GPT5.2

Results and Insights

Longer notes → lower performance



F1 performance on the Dev set decreases as note length increases, ~10% drop from 10 to 50 sentences.

Clinical notes contain multiple stages of care (finding, decision, procedure).
Evidence alignment must reflect clinical meaning, not just lexical or semantic overlap.

RESULTS

77.4%

F1 overall (Test)

Zero-shot with clinical reasoning

+2%

F1 overall (Dev)

Improvement with clinical reasoning

Scan for the code



Mengxuan



Nicolay

